

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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Anonymous submission

ABSTRACT

Representation alignment (REPA) has become a broadly adopted way to guide generative training, distilling representations from a strong, pretrained vision encoder into intermediate diffusion features. Across the many encoders and settings in which this paradigm is applied, we ask a fundamental question that cuts across them all: what aspect of the target representation actually matters for generation, its *global* semantic information (e.g., as measured by ImageNet-1K accuracy) or its *spatial* structure (i.e. pairwise cosine similarity between patch tokens)? The prevailing wisdom across the field holds that stronger global semantic performance is what makes a representation a better alignment target. To probe this assumption at scale, we conduct a large-scale empirical analysis spanning 27 diverse vision encoders and a range of model scales. The results are surprising and remarkably consistent — spatial structure, rather than global performance, is what drives the generation performance of a target representation across these settings. Motivated by this general principle, we introduce two straightforward, broadly applicable modifications that specifically accentuate the transfer of *spatial* information. We replace the standard MLP projection layer in REPA with a simple convolution layer and introduce a spatial normalization layer for the external representation. Despite its simplicity (implemented in <4 lines of code), our method, termed iREPA, consistently improves the convergence speed of REPA across a wide spectrum of vision encoders, model sizes, resolutions, and training paradigms (including REPA, REPA-E, MeanFlow, JiT, and beyond). More broadly, our work motivates revisiting the fundamental working mechanism of representation alignment and rethinking how it can be leveraged for improved training of generative models in general.

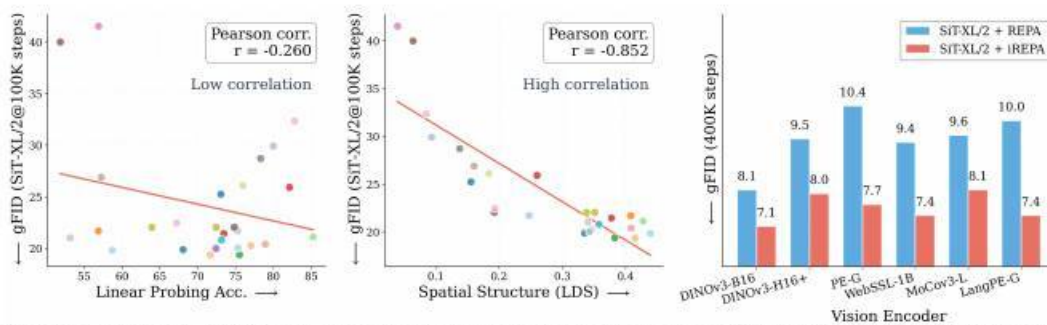


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis spanning 27 diverse vision encoders. Surprisingly, and contrary to the prevailing wisdom, we find that spatial structure —rather than global performance (measured by linear probing accuracy)— is the general driver of the generation performance of a target representation. *Right:* Building on this, we introduce two simple modifications that accentuate the transfer of spatial features from the target representation to the diffusion model. This simple, recipe-agnostic approach consistently improves the convergence speed of REPA across a broad range of settings.



Figure 2: **Motivating examples — spatial structure matters.** Comparisons drawn from different encoder families, all showing the same inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite a significantly higher validation accuracy (82.8% vs. 53.1%), yields worse generation than SpatialPE-B (Bolya et al., 2025). *Right:* Likewise, WebSSL-1B (Fan et al., 2025) has much better global performance (76.0% vs. 53.1%) yet worse generation. **Spatial Self-Similarity:** Across these families, spatial structure —rather than global performance— provides the better predictor of generation quality. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

1 INTRODUCTION

Representation alignment has emerged as a general and powerful technique for accelerating the training of diffusion transformers and, more broadly, modern generative models (Ma et al., 2024; Peebles & Xie, 2023). By aligning internal diffusion representations with pretrained self-supervised visual encoders, a growing family of methods has demonstrated significant improvements in convergence speed and final performance across many settings (Yu et al., 2024; Leng et al., 2025a). Yet, despite these wide-ranging empirical successes, there remains limited understanding of the precise mechanisms through which self-supervised features enhance generative training. This leaves open a fundamental question that applies to any choice of encoder or model: is the improvement primarily driven by incorporating better global semantic information, as commonly measured through linear probing performance, or does it instead stem from better capturing spatial structure, characterized by the relationships between patch token representations?

Understanding these mechanisms is crucial for advancing generative model training in general, as it directly shapes how practitioners select target representations and maximize their benefits across the full range of available encoders. Currently, a widely held assumption across the field is that an encoder’s suitability for representation alignment correlates strongly with ImageNet-1K validation accuracy, a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). That is, target representations with better ImageNet performance are, as a general rule, hypothesized to lead to better generation (Yu et al., 2024). As such, increases in linear probing accuracy of diffusion features are routinely cited as evidence for the effectiveness of representation alignment, reinforcing a broad emphasis on global semantic information as the primary driver of improvement. The following quote from REPA (Yu et al., 2024) captures this prevailing understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we challenge this conventional wisdom by systematically investigating what truly drives representation alignment in general: global semantic information or spatial structure? Through large-scale empirical analysis spanning a broad and representative range of vision encoders —from classical hand-crafted descriptors to recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024)— we uncover 3 surprising findings that hold consistently across these settings.

Higher validation accuracy does not imply a better representation for generation. Contrary to prevailing assumptions, and consistently across the encoders we study, vision encoders with higher global semantic performance —measured by ImageNet-1K linear probing accuracy— can often underperform for generation. For instance, consider PE-Spatial-B, a spatially-tuned model derived from PE-Core-G (Bolya et al., 2025). We find that while PE-Spatial-B shows a much worse validation

accuracy¹ on patch tokens than PE-Core-G (53.1% vs. 82.8%), it leads to better generation with REPA (Figure 2). Similarly, WebSSL-1B (Fan et al., 2025) shows much better global performance (76.0% vs. 53.1%), yet worse generation. In Section §2, we show that this pattern is not an isolated artifact but recurs across a wide variety of target representations, pointing to a general principle governing how representation alignment benefits generative training.

Spatial structure, not global information, determines generation performance. To quantify this insight in a way that generalizes across encoders, we consider several straightforward metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) between patch tokens (§3). We then perform a large-scale analysis computing the correlation between generation FID with REPA, linear probing accuracy, and spatial structure across 27 vision encoders and 3 model sizes (SiT-B, SiT-L, SiT-XL). Strikingly, and uniformly across these settings, all spatial structure metrics exhibit far stronger correlation (Pearson $|r| > 0.852$) with generation FID scores than the widely used ImageNet-1K validation accuracy ($|r| = 0.26$). This general principle is reinforced by additional experiments showing that even external representations with very limited global information yield significant gains with REPA. For instance, SAM2 leads to better generation with REPA than other encoders carrying $\sim 60\%$ higher ImageNet accuracy (Fig. 3). More strikingly still, we find that classical spatial features such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005)—though not the strongest targets—can also deliver meaningful gains with representation alignment (§3), underscoring how broadly the principle applies beyond modern neural encoders.

Accentuating spatial features improves convergence performance. Building on this general principle, we next introduce two straightforward modifications (<4 lines of code) that specifically accentuate the transfer of “spatial” information from the target representation to the diffusion model, and that are agnostic to the particular encoder or training recipe used: (1) We first introduce a spatial regularization layer that boosts the spatial contrast of the target representations. (2) Next, we show that the standard MLP-based projection layer (used to map diffusion features to target representation dimensions) causes a loss of spatial information (Figure 6a); to avoid this, we replace it with a simple convolution layer. The results are surprising: our simple method, termed iREPA, consistently improves the convergence speed of REPA across a wide range of encoders, model sizes, and training recipes, *e.g.*, REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) w/ REPA.

We highlight the main contributions of this paper below:

- We perform a large-scale empirical analysis showing that, as a general principle, spatial structure—and not global semantic information—drives the effectiveness of representation alignment.
- We introduce a straightforward and fast-to-compute Spatial Structure Metric (SSM) that shows significantly higher correlation with downstream FID performance than linear probing scores, robustly across encoders, model scales, and evaluation metrics.
- We propose simple, recipe-agnostic modifications that better accentuate the transfer of spatial information from the target representation to diffusion features. Our method delivers consistent convergence-speed improvements over REPA across a broad spectrum of target representations, model architectures, resolutions, and training-recipe variants (REPA, REPA-E, MeanFlow w/ REPA, JiT w/ REPA, and beyond).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We first provide several motivating examples, drawn from a broad range of encoder families, showing that a target representation with higher global performance (ImageNet-1K accuracy) does not imply better generation performance with REPA. We later show in §3 that these previously-unexplained observations—which span diverse encoders, model families, and manipulations—can instead be uniformly explained by measuring the spatial structure of the target representation.

Recent vision encoders. We examine a broad selection of recent vision encoders spanning distinct design philosophies, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As seen in Figure 2, we find that while

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure the global semantic performance of an external representation, as only patch tokens are used for representation alignment.

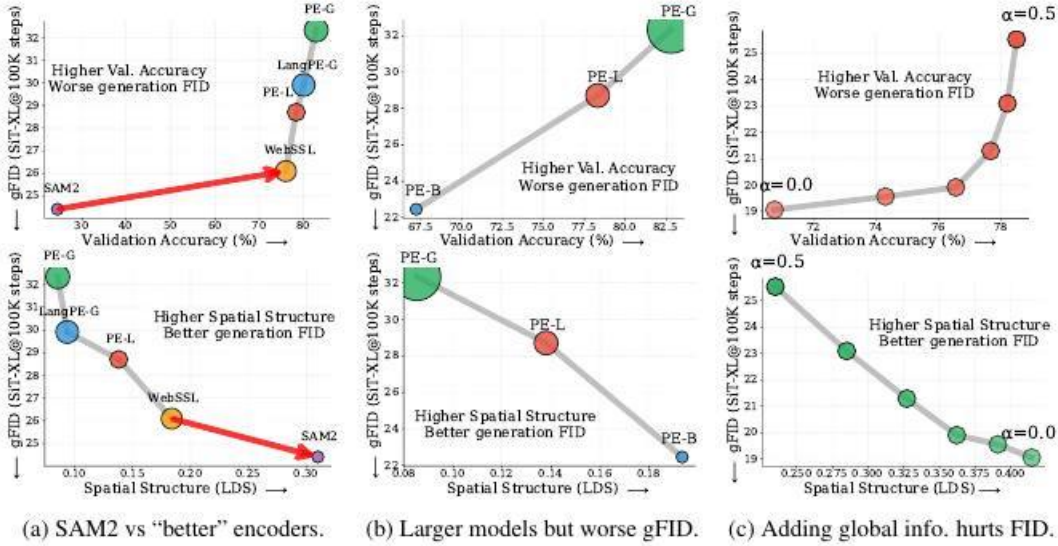


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** A collection of trends, drawn from distinct settings, all showing that global performance is a poor predictor of generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, yields better generation with REPA than other models carrying $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can have better validation but worse generation. (c) Adding global information to patch tokens via the CLS token improves global performance yet hurts generation. **Bottom row:** Across all of these cases, spatial structure—rather than global performance—provides a consistently better indicator for generation. See §3 for a large-scale, detailed analysis across several spatial structure metrics. All results are reported w/o classifier free guidance, SiTXL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%), it performs worse as the target representation for REPA (FID 32.3 vs 21.0). The same reversal holds for a very different encoder family: WebSSL-1B (1.2B) achieves much higher ImageNet-1K accuracy (76.0% vs 53.1%) yet still performs worse as a target representation for REPA (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To probe the extremes of this phenomenon—where global information is almost entirely absent—we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024), a small model carrying very little global information, with a validation accuracy of only 24.1% (Fig. 3a). Surprisingly, when used for REPA, SAM2-S achieves better FID than vision encoders with dramatically higher ImageNet-1K accuracy, *e.g.*, PE-Core-G (82.8%). This reinforces that the trend is not confined to a narrow accuracy band but persists even at the low-global-information end of the spectrum.

Larger models within the same encoder family can have similar or worse generation performance. A common perception is that scaling up a model within an encoder family reliably yields better representations (as measured by ImageNet-1K accuracy). However, for representation alignment, this scaling intuition breaks down across multiple families: larger variants often lead to similar (DINOv2) or even *worse* (PE, C-RADIO) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a similar observation for DINOv2 and explain it as “we hypothesize is due to all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. We later show that this trend, across families, is better explained through spatial structure (§3).

Adding more global information can hurt generation. Beyond comparing existing encoders, we can test the underlying principle directly through a controlled intervention that injects global semantics, via the CLS token, into local patch tokens (using DINOv2 as encoder). The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As shown in Figure 3c, as α increases from 0 to 0.5, linear probing accuracy improves monotonically from 70.7% to 78.5%. Yet generation quality deteriorates substantially, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$ —demonstrating causally, and not merely correlationally, that more global information can be actively harmful.

Taken together, these observations—spanning distinct encoder families, model scales, and a direct causal intervention—consistently indicate that the global performance of a representation is not a

reliable indicator of its REPA performance. We next show (§3) that spatial structure instead provides a broadly predictive signal.

- **Finding 1.** *As a general rule, better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.*

3 SPATIAL STRUCTURE MATTERS MORE

We hypothesize that spatial structure, rather than global information, is the general driver of generation performance for a target representation. To quantify this in an encoder-agnostic way, we first consider several straightforward metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then show that, across the board, all of these spatial structure metrics not only correlate far more strongly with gFID than ImageNet-1K accuracy, but also unify and explain the previously unexplained observations of §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then measure *spatial* self-similarity metric as how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization (closer patches more similar to each other than patches further away). By default, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed as the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, and find the resulting correlation to be robust to the exact choice of these radii. Please see Appendix B for exact details and for a family of alternative spatial metrics (CDS, SRSS, RMSC) that behave equally effectively (see Fig. 4) —indicating that our conclusions do not hinge on any single metric definition.

Spatial structure correlates much higher with generation performance than linear probing. We next perform a large-scale correlation analysis spanning 27 diverse vision encoders, chosen to cover a wide range of architectures, training objectives, and scales. As shown in Figure 4, the commonly used linear probing accuracy shows very weak correlation with generation quality (Pearson $|r| = 0.26$), whereas all SSM metrics show much stronger correlation (Pearson $|r| > 0.85$) —a gap that holds consistently across this broad encoder pool.

Generalization across model scales. This behaviour is not specific to a single model size. We verify the correlation across model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing remains weakly correlated across all scales ($|r| < 0.306$), while spatial structure remains strongly correlated with generation performance throughout ($|r| > 0.826$), indicating that the principle generalizes across model capacities.

- **Finding 2.** *Across encoders and model scales, spatial structure correlates much more strongly with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends under a single principle.*

Spatial structure can explain previously unexplained trends. As discussed in §2, global performance (ImageNet-1K accuracy) does not serve as a predictive measure of effectiveness for representation alignment. Instead, the spatial metrics above provide a single, unifying explanation for the full set of otherwise disparate observations.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%), but performs worse as a target representation for REPA (FID 32.3

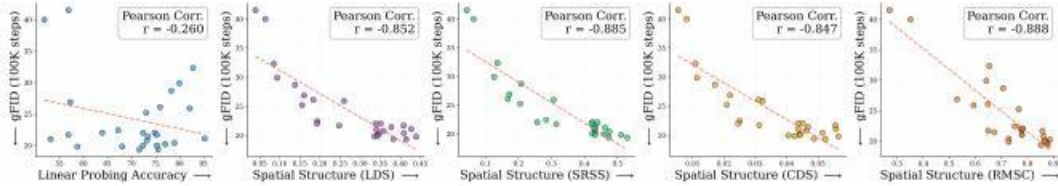


Figure 4: **Spatial structure shows higher correlation with generation quality than linear probing.** Correlation analysis spanning 27 diverse vision encoders, with SiT-XL/2 and REPA. Linear probing correlates only weakly with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric —LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)— correlates far more strongly with generation performance, and does so consistently across this broad encoder pool. See Fig. 10 for detailed plots with encoder labels.

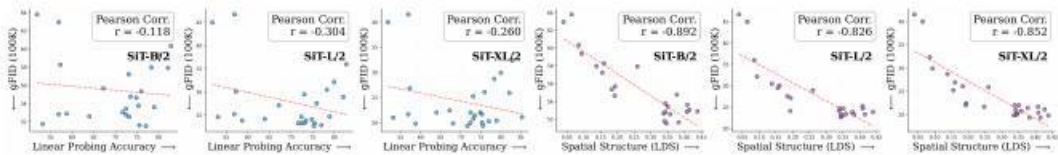


Figure 5: **Correlation analysis across model scales.** At every model scale considered, spatial structure (right) consistently correlates more strongly with gFID than linear probing (left) —indicating that the effect generalizes across model capacities rather than being specific to any one size.

vs 22.0). The spatial structure metric resolves the puzzle: despite its lower global performance, PE-Spatial-B exhibits much stronger spatial structure than PE-Core-G, which accounts for its better generation performance.

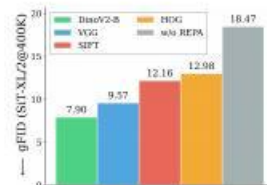
(2) *SAM2 outperforms “better” vision encoders:* §2 shows that although SAM2 achieves much lower ImageNet-1K accuracy (24.1%), it leads to better generation than encoders with $\sim 60\%$ higher accuracy. As in Fig. 3a, these gains are directly explained by SAM2’s stronger spatial structure —the same explanation that applies to the other encoders.

(3) *Larger models in the same encoder family underperform:* As shown in Fig. 3b, larger models within an encoder family attain better ImageNet-1K accuracy yet can carry weaker spatial structure, and thus worse generation performance with REPA. This is consistent with the observations of Yu et al. (2024), who find that larger models can yield similar or worse generation performance.

4) *Adding global information to patch tokens via the CLS token hurts generation:* In Figure 3c, increasing global information by mixing the CLS token into patch tokens improves global performance, but reduces spatial contrast among patch tokens. Tokens then become highly similar to otherwise unrelated ones (e.g., foreground object matching background). The resulting loss of spatial structure again explains the drop in generation performance, closing the loop under a single principle.

If spatial structure matters more, can we use SIFT or HOG features for REPA?

Surprisingly, yes. Even classical, non-learned spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), as well as intermediate VGG features (Simonyan & Zisserman, 2014), all yield performance gains with REPA —though, as expected, they are not the strongest targets. This substantially widens the scope of what can serve as a useful alignment target, providing further evidence that representation alignment benefits from spatial features alone, without any need for additional global information.



Can we use spatial metrics to explain gains with REPA? Yes. As shown, the introduced spatial metrics explain gains not only with standard REPA but also with our improved training recipe (iREPA), introduced next in §4 —suggesting they capture a general property that transfers across training recipes.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

To further investigate the role of spatial structure in representation alignment, we introduce two straightforward and broadly applicable modifications to the original REPA training recipe that enhance the transfer of spatial features from the teacher (vision encoder) to the student (diffusion transformer) model. Both modifications are deliberately generic—they make no assumptions about the specific encoder, model size, or training recipe, and so are expected to transfer across the many settings in which representation alignment is used.

Convolutional projection layer instead of MLP. The standard REPA approach uses a 3-layer MLP projection to map diffusion feature dimensions to those of the external representation. However, as shown in Figure 6a, this projection is lossy and diminishes the spatial contrast between patch tokens—a limitation that is inherent to the MLP design rather than tied to any particular encoder. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid. The convolutional structure naturally preserves local spatial relationships through its inductive bias, and does so regardless of the target representation used.

Spatial normalization layer. Consistent with Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders generally contain a significant global component. This is evidenced by the high linear probing scores obtained from the mean of patch tokens (Figure 14), a pattern that recurs across the encoders we examine. Furthermore, Figure 6b shows that while mixing this global information (mean of patch tokens) into the patch tokens improves global performance, it reduces the spatial contrast between individual patch tokens, causing tokens (e.g., a foreground object) to appear highly similar to otherwise unrelated tokens (e.g., background).

Given the general principle established in §3, we hypothesize that this global information (mean of patch tokens) can be safely sacrificed to improve the spatial contrast between patch tokens across encoders. The improved contrast should yield a cleaner spatial signal (pairwise similarity between patch tokens) and, in turn, better REPA performance. To this end, we add a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

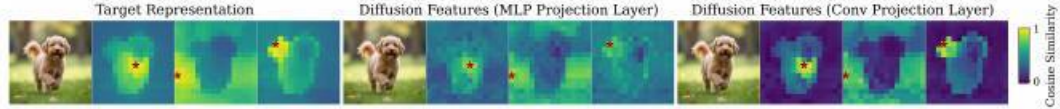
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# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
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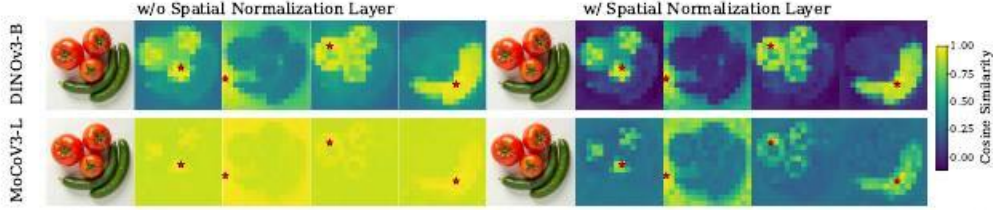
4.1 EXPERIMENTS

In this section, we validate the improved training recipe through extensive experiments, using ImageNet 256×256 as our primary testbed. In particular, we investigate the following research questions, each aimed at establishing the generality of the approach:

- Does iREPA consistently improve the convergence speed of diffusion transformers over REPA across a wide range of external representations? (Figure 7, 8, 12, 13, 15)
- Is iREPA scalable in model size and robust across variations in training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and Figure 12, 16)
- Does iREPA generalize to more recent representation alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), as well as to pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])



(a) *Simpler projection layer for REPA.* The standard MLP projection layer in REPA (middle) loses spatial information when transferring features from the target representation (left) to the diffusion features. Replacing it with a simpler convolution layer yields better spatial information transfer (right), independent of the encoder used.



(b) *Spatial normalization layer.* The patch tokens of pretrained vision encoders generally carry a global component that limits spatial contrast, causing tokens in one semantic region (e.g., tomato) to show non-trivial cosine similarity with unrelated tokens (e.g., background or cucumber). We hypothesize that this global information can be sacrificed to improve the spatial contrast between patch tokens, leading to better generation performance —across encoders.



(c) *Overall impact of improved training recipe (iREPA) on spatial structure of diffusion features.*

Figure 6: Motivating two straightforward, encoder-agnostic modifications (iREPA) that enhance spatial feature transfer from the target representation to diffusion features. All results reported with SiT-L/2 and REPA at 100K.

Convergence Speed. We evaluate the convergence behavior of iREPA over a broad set of vision encoders spanning distinct training paradigms (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and multiple model sizes (SiT-XL/2 and SiT-B/2). Results are shown in Fig. 7. Across every combination of target representation and model size we consider, iREPA consistently accelerates convergence relative to baseline REPA —indicating that the benefit is a general property rather than an encoder- or scale-specific effect.

Target representation. We analyze the generalization of iREPA over a wide variety of vision encoders in Fig. 8 and Table 4. iREPA consistently improves generation quality across all vision encoders considered, without exception. Comprehensive comparisons spanning the full set of 27 encoders are provided in Appendix C, E, further underscoring the breadth of encoders for which the method holds.

Enc. Size	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

Model Size	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

Aln. Depth	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA yields consistent gains over baseline REPA across a broad range of training settings, spanning encoder size, model size, and alignment depth. Unless otherwise specified, all results use DINOv2 as the encoder, SiT-XL/2, 100K steps, and vanilla-REPA as the baseline.

Encoder size. We next examine how iREPA behaves as encoder capacity is scaled. Table 1a reports results spanning the full PE family across nearly two orders of magnitude in size —PE-B (90M), PE-L (320M), and PE-G (1.88B). Across this entire range, iREPA consistently improves performance.

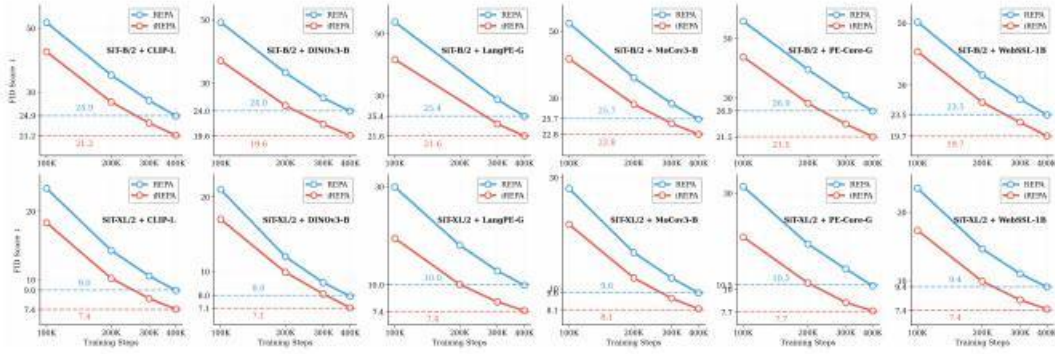


Figure 7: Accentuating spatial features consistently improves convergence speed across settings.

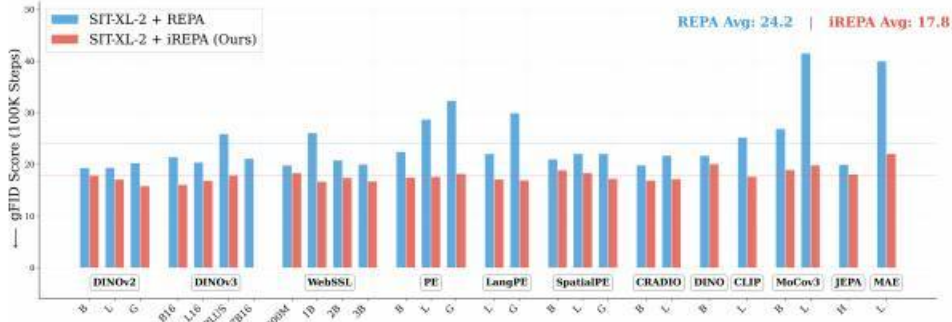


Figure 8: **Variation in target representation.** Across the entire set of 27 vision encoders—spanning diverse architectures and training objectives—accentuating the transfer of spatial features from the target representation to diffusion features (iREPA) yields consistent improvements in convergence speed, with no exceptions. See Appendix C for further results across a broad range of settings.

Notably, the relative improvement grows with encoder size (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G), indicating that the benefit not only persists but strengthens for larger encoders.

Scalability. We analyze the scalability of iREPA across model scales in Table 1b. The spatial improvements not only hold consistently, but yield larger relative gains as model size increases—demonstrating that the approach scales gracefully with model capacity rather than saturating.

Encoder depth. Table 1c analyzes the generalization of iREPA across alignment depths. All results use SiT-B/2 at 100K iterations with DINOv3-B as the target representation. Improvements over baseline REPA are consistent at every alignment depth, showing the method is not sensitive to this design choice.

Ablation on different components. We also study the role of the individual components of iREPA in Table 2. Across the different encoders tested, both the spatial normalization and the convolution projection layer independently improve generation quality over baseline REPA, with the best results consistently achieved by combining the two—indicating the two components are complementary in general, not just for a single encoder.

Training recipe. We further analyze the generalization of iREPA beyond standard REPA, to more recent training recipes such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows that the spatial improvements carry over to these distinct recipes, yielding convergence gains in each case—evidence that the benefit is a recipe-agnostic property of accentuating spatial transfer.

Classifier-free guidance. We evaluate the generation quality of iREPA with CFG in Table 4, which presents results across a range of vision encoders at 400K training iterations. Across all encoders,

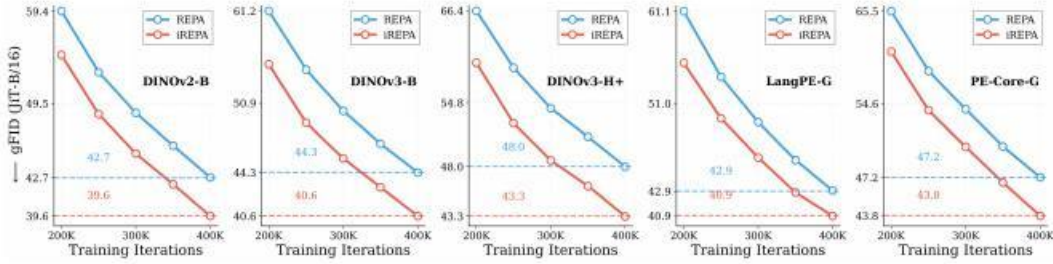


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA with pixel-space JiT (Li & He, 2025b) across a range of vision encoders. Even in the pixel-space regime, accentuating the transfer of spatial structure consistently improves convergence speed across encoders —showing the benefit is not confined to latent-space diffusion. All results are reported with JiT-B/16 (Li & He, 2025b), a batch size of 256, and without classifier-free guidance. Refer to Table 10 for further results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across every encoder considered, both the spatial normalization layer and the convolution projection layer (§4) independently deliver significant convergence-speed gains, with the best results consistently obtained by combining the two. All results reported at 100K steps with SiT-XL/2.

iREPA leads to faster convergence both with and without classifier-free guidance, confirming that the gains are not an artifact of a particular inference-time setting.

Pixel-space diffusion (JiT). To test whether the approach extends beyond latent-space diffusion, we also evaluate iREPA on pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA using JiT (Li & He, 2025b). Even in this different modeling regime, accentuating the transfer of spatial information consistently achieves faster convergence with REPA across a variety of vision encoders (e.g., DINOv2, DINOv3, PE, etc.), indicating that the principle is not tied to operating in latent space.

- **Finding 3.** *Accentuating the transfer of spatial structure from the target representation to diffusion features boosts convergence speed with representation alignment (REPA), consistently across encoders, model sizes, resolutions, and training recipes.*

5 RELATED WORK

We discuss the most relevant related work here and provide a detailed discussion in Appendix H.

Representation alignment for generation. A large and rapidly growing body of work explores the use of external representations for improving diffusion model training (Pernias et al., 2023; Fuest et al., 2024). Notably, recent works (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) show that significant performance gains can be achieved by aligning internal diffusion features with clean image features from a pretrained vision encoder. The paradigm has already reached well beyond image synthesis: (Zhang et al., 2025; Wu et al., 2025) extend it to video and 3D generation respectively, and (Ma et al., 2025a) shows that representation alignment can improve the training of unified models. This breadth of application makes the underlying mechanism all the more important, yet it remains poorly understood across these settings. In this paper, we ask what aspect of the target representation actually matters for generation, and turn the answer into a broadly applicable, improved training recipe.

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Layering our spatial improvements on top of two distinct, more recent recipes —REPA-E (a) and MeanFlow with REPA (b)— yields consistent gains in each case, indicating the benefit transfers across training recipes rather than being specific to standard REPA. For MeanFlow, we report results at 1 and 4 NFEs, using a CFG value of 2.0. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across a range of encoders, with and without classifier-free guidance.** SiT-XL/2 is used as the base model, and the gains hold consistently in both guidance settings. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across the full encoder set.

Spatial vs global information tradeoff in pretrained vision encoders. A number of recent works examine the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued training of self-supervised vision representations can increase the similarity between the global CLS token and patch tokens, degrading performance on dense spatial tasks, while (Bolya et al., 2025; Heinrich et al., 2024) deliberately train spatially-tuned models for dense spatial tasks. Our findings connect this line of work to generative modeling: for generation, the spatial structure of a vision encoder matters more than its global information—a conclusion that holds across the diverse encoders we study. We hope this motivates broader future research on how to select, evaluate, and train external representations for generation in general.

6 CONCLUSION

In this paper, we ask what truly drives the effectiveness of representation alignment: the global information or the spatial structure of the target representation? Through a large-scale empirical analysis spanning many encoders and model scales, we arrive at a surprising and broadly consistent finding: spatial structure, and not global information, is what drives the effectiveness of representation alignment. Building on this general principle, we introduce two simple, recipe-agnostic modifications that accentuate the transfer of spatial information from the target representation to diffusion features. Our method, termed iREPA, consistently improves convergence speed with REPA across a wide spectrum of vision encoders, model sizes, resolutions, and training recipes. More broadly, we hope this work motivates the community to revisit the fundamental working mechanism of representation

594 alignment and to rethink how it can be leveraged for improved training of generative models across
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REPRODUCIBILITY STATEMENT

We provide complete implementation details and hyperparameters, shared across all encoders, model sizes, and training recipes, in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance to facilitate reproduction across the full range of settings we study.

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A DETAILED CORRELATION ANALYSIS

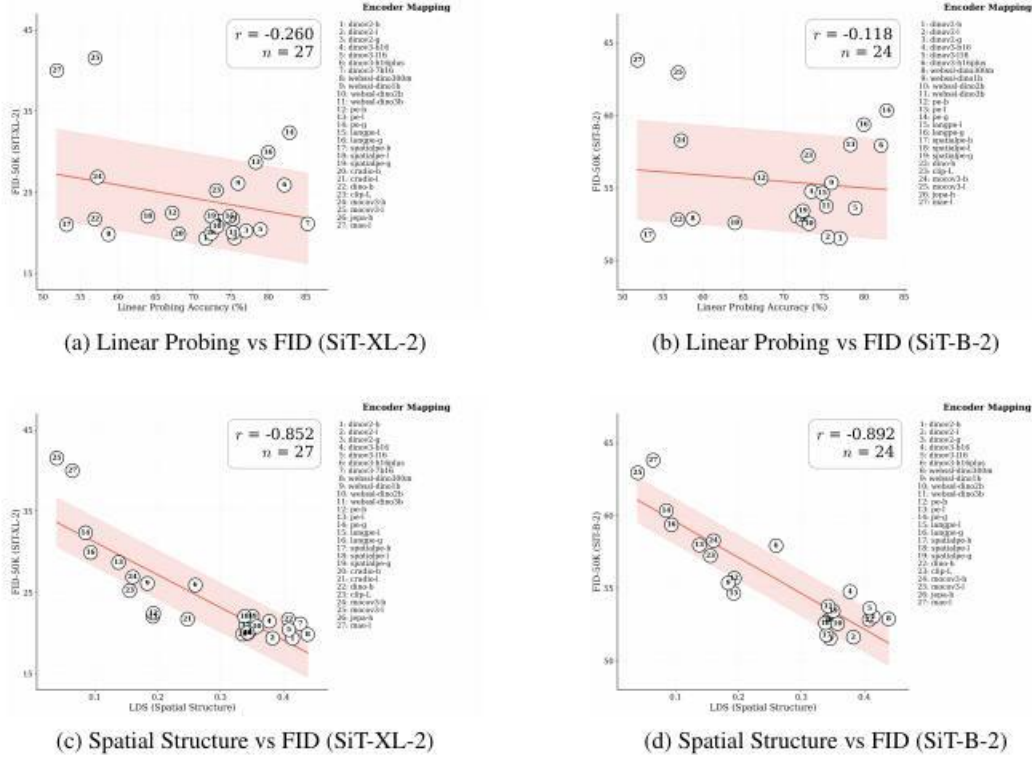


Figure 10: Detailed correlation analysis with encoder labels. The top row shows the linear probing accuracy vs FID correlation for the SiT-XL-2 and SiT-B-2 models, and the bottom row shows the spatial structure (LDS) vs FID correlation. Each point is labeled with its corresponding encoder number (see legend). Across both model scales—and across the full pool of encoders—the spatial structure metric demonstrates a consistently and significantly stronger correlation with generation quality.

B SPATIAL SELF-SIMILARITY METRICS

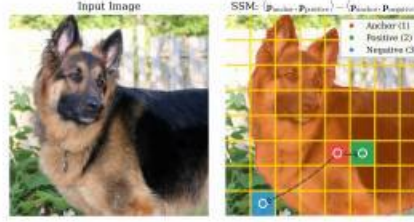


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of the SRSS metric: for each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within the mask) and anchor-negative pairs (outside the mask). Intuitively, larger values indicate that patches in the same semantic region are more similar than patches in unrelated regions, signaling better spatial structure preservation. The construction is generic and applies uniformly to any encoder’s patch features.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We next discuss several straightforward, encoder-agnostic metrics for measuring the spatial self-similarity structure of target representations. Each captures a complementary facet of spatial organization, and, as we show, they lead to consistent conclusions across the full range of encoders considered.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are the hyperparameters. By default we use $r_{\text{near}} = r_{\text{far}} = H/2$. We found correlation to be robust to their exact choices. Intuitively, larger values indicate that *on average*, patches that are closer to each other are more similar than patches that are further away.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance, suggesting stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsampled to $H \times W$. We then select three types of points: anchor, positive, and negative. The anchor and positive points are sampled from within the mask (should have similar semantics), while the negative point is sampled

from outside the mask (less related to anchor). Conceptually, if the encoder feature representation preserves the spatial structure, the anchor and positive points should have higher cosine similarity, while the anchor and negative points should have lower cosine similarity. Thus, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

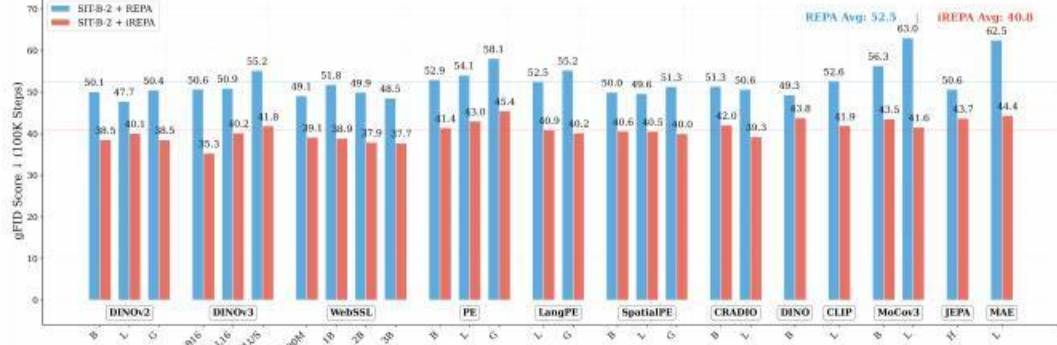
where positive points are within Manhattan distance $d \leq r_{\text{near}}$ from the anchor within the mask, and negative points are at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric that captures the spatial contrast between patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

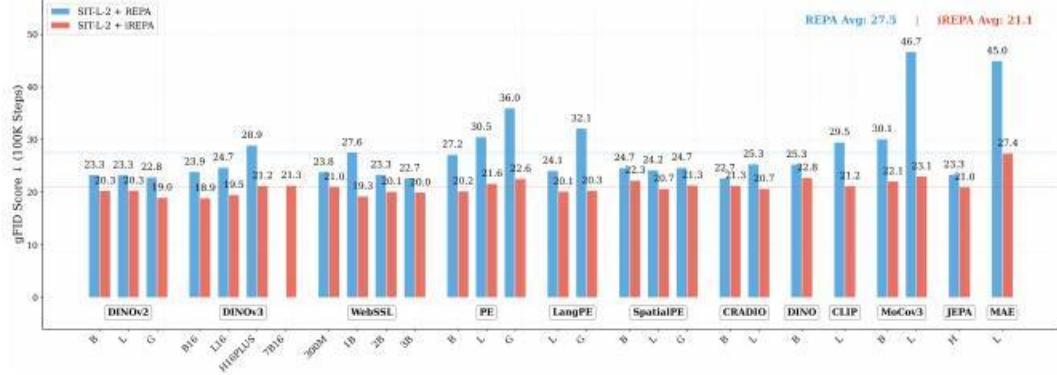
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations, suggesting preserved spatial structure, whereas lower values indicate more uniform feature distributions and hence a loss of spatial structure—a regime that typically arises when global information comes to dominate and suppresses spatial structure (Siméoni et al., 2025).

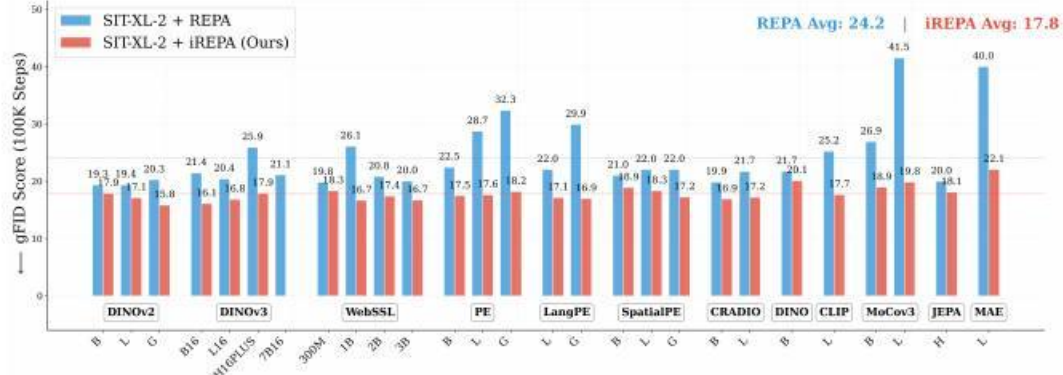
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



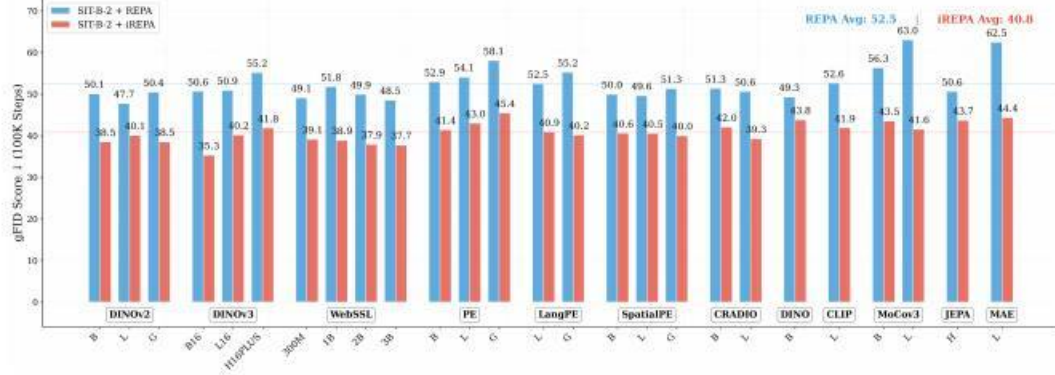
(b) SiT-L-2 (458M parameters)



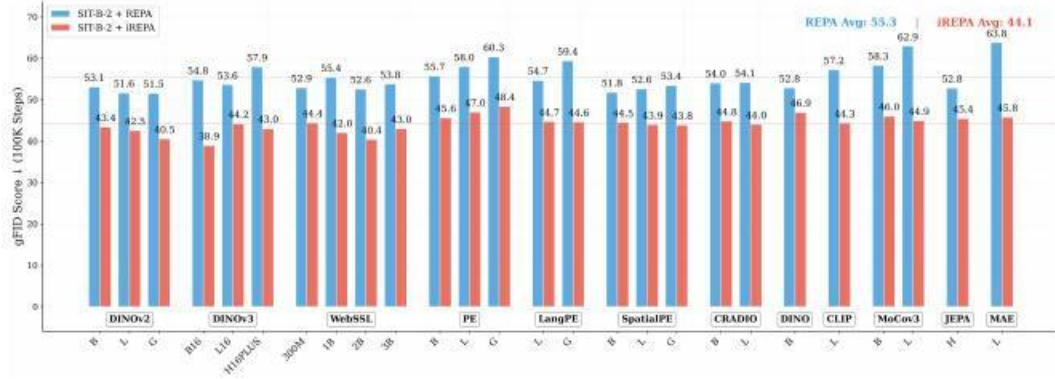
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements over a wide range of vision encoders for the SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across every encoder and model size considered.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance at different encoder depths (6 vs 8 layers) across a range of vision encoders. The improvements from iREPA hold consistently regardless of encoder depth, with slightly stronger performance at 8 encoder layers. Both configurations show substantial FID improvements once spatial regularization is applied, indicating the gains are robust to this design choice.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: SiT-B/2 performance across a broad set of vision encoders at 400K iterations. iREPA consistently improves generation quality for every encoder in this diverse pool, with no exceptions. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across a wide range of vision encoders at 400K iterations.** iREPA consistently improves generation quality across every encoder considered, with particularly strong gains in the FID and IS metrics —mirroring the trends seen at other model scales. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

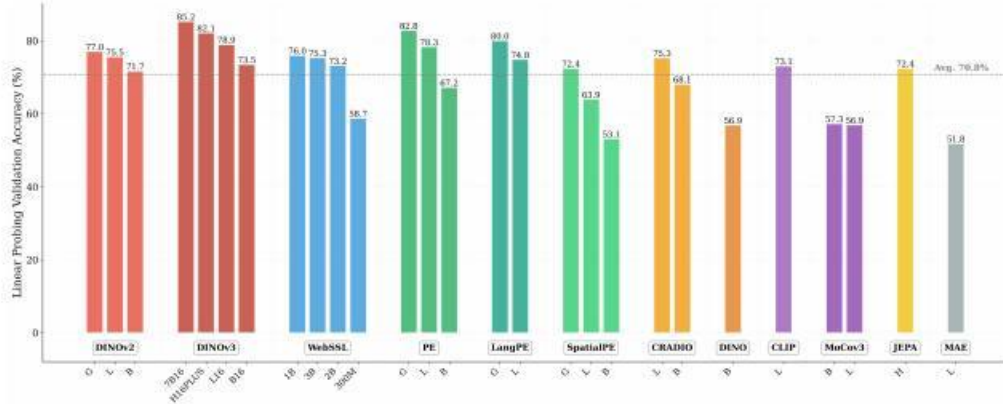


Figure 14: **Global information in mean patch tokens.** Across the encoders we examine, we find that beyond the [CLS] token, the mean of patch tokens also carries substantial global semantic information. While this boosts global performance, it reduces the contrast between individual patch tokens and can thereby hinder spatial structure transfer. This motivates removing part of this global information (via the mean of patch tokens) to improve spatial structure transfer during representation alignment (§4) —a remedy that applies generally rather than to any single encoder.

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA and iREPA with CFG (scale 2.0) across a range of vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both the IS and FID metrics for every encoder considered.

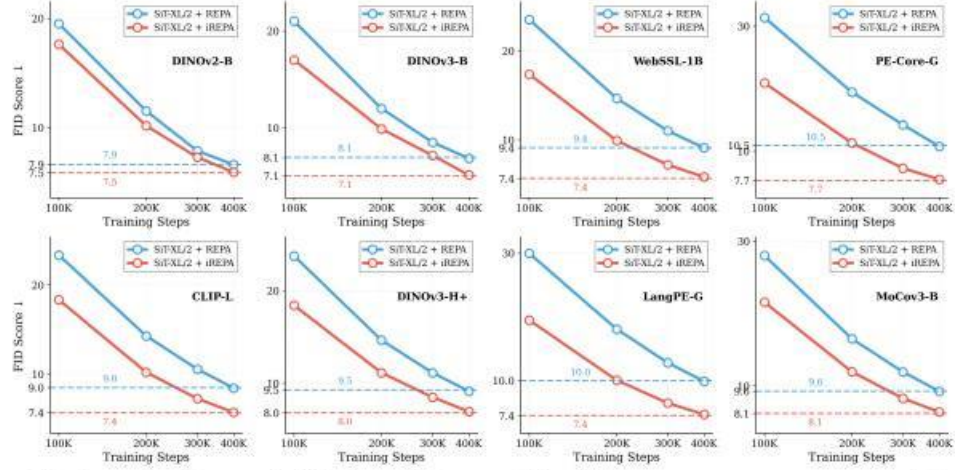


Figure 15: Accentuating spatial features helps consistently improve convergence speed. Results for SiT-XL/2 with REPA and iREPA.

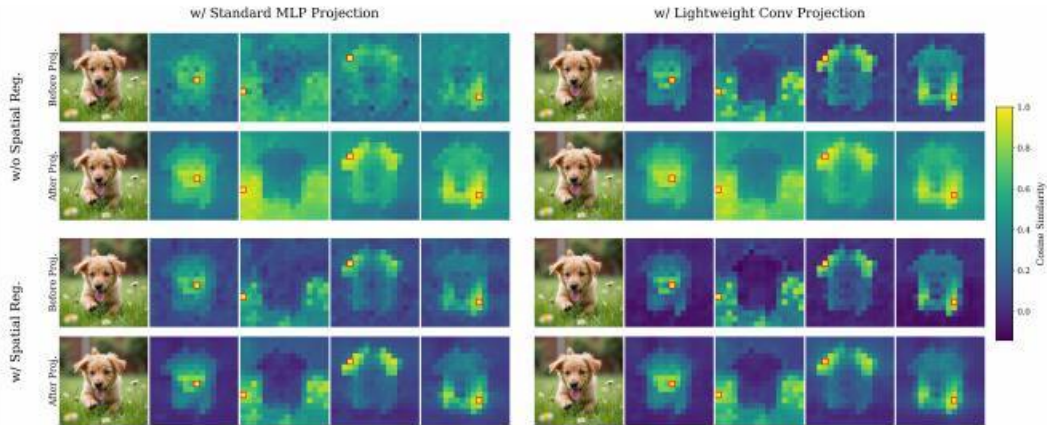


Figure 16: Visualizing the impact of two straightforward, broadly applicable improvements that enhance spatial feature transfer. First, the standard MLP projection layer (top-left) loses spatial information while transferring features from the pretrained encoder (after projection) to the diffusion model (before projection). Replacing it with a simpler convolution layer better preserves the spatial information transfer (top-right). Second, vision encoder features often exhibit limited spatial contrast (§2), which causes tokens in one semantic region (e.g., dog) to show non-trivial cosine similarity with unrelated tokens (e.g., background). We address this with a simple spatial regularization layer that accentuates the spatial contrast in the learned representation, leading to better generation performance. As a general rule, the best results are obtained by using both components together (refer to Table 2).

G IMPLEMENTATION DETAILS

Training setup. To keep comparisons controlled and directly comparable across settings, we follow the same setup as REPA (Yu et al., 2024) unless otherwise specified. Our primary experiments train on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol from ADM (Dhariwal & Nichol, 2021), where each image is center-cropped and resized to 256×256 resolution. We use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout diffusion training and inference. For the spatial normalization layer we use $\gamma \in [0.6, 0.8]$, and for the projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` for acceleration, together with gradient clipping and an exponential moving average (EMA) of the generative model for stable optimization. These choices are shared unchanged across the many encoders, model sizes, and recipes we study, so that the observed gains can be attributed to the method rather than to setting-specific tuning.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments, we obtained the implementation (including REPA) from the original authors and adapted it by introducing the same two straightforward iREPA changes (§4). In each case, we apply the identical modifications without recipe-specific redesign, which is what allows the method to slot into such different training pipelines.

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using the Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), all measured on 50K generated images; this full suite of complementary metrics is used so that conclusions do not rest on any single measure. For sampling, we follow REPA (Yu et al., 2024), using the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) provide robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has focused almost exclusively on representations from modern self-supervised methods with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings suggest a broader perspective: because spatial structure preservation is what matters for generation quality, even classical spatial features can improve diffusion training when properly aligned. This considerably widens the space of viable alignment targets, spanning the full spectrum of spatial feature extractors—from traditional handcrafted descriptors to modern learned representations—provided they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models across a wide range of roles: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) using CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, yielding clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, showing that representation alignment can improve both generation convergence and quality. Cutting across these varied uses, our work argues that what matters is the spatial structure such encoders preserve, rather than their discriminative capabilities.

Denoising transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, across a range of formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures such as GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis, U-ViT (Bao et al., 2023) shows that pure transformer backbones without U-Net inductive biases can be competitive, and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation for flexible quality-speed trade-offs. Despite these architectural advances, training such models from scratch remains computationally expensive, often requiring millions of iterations to reach good generation quality. While representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanism behind this improvement has remained unclear. Our analysis identifies a common underlying cause: the key benefit comes from preserving spatial structure rather than semantic alignment—which explains why certain encoders provide stronger acceleration than others across architectures, and points toward more effective alignment strategies in general. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. Collectively, these works establish denoising as a fundamental self-supervised task that naturally encourages the learning of robust features. Our findings complement this view: when diffusion models are aligned with strong spatial representations from self-supervised encoders, both generative and discriminative capabilities improve, pointing to spatial structure preservation as a general factor underlying this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets, across a broad set of training objectives. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion unlabeled images by aligning representations from different augmentations. WebSSL (Fan et al., 2025) shows that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) distills multiple teacher models into versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space rather than reconstructing pixels, SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training, and the Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction. Although these models are typically evaluated on global tasks such as image classification, our work reveals a crucial insight that spans this entire landscape: strong performance on discriminative benchmarks does not necessarily translate into better generation quality. Instead, across objectives and scales, encoders that preserve spatial structure—regardless of their classification accuracy—provide the most benefit for diffusion training. This argues that the evaluation of self-supervised encoders should be reconsidered whenever the target is a generative application, treating spatial coherence as at least as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 ([OpenAI, 2025](#)) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code ([Anthropic, 2025](#)). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, invoking the increase in validation accuracy to explain the effectiveness of REPA. We find that spatial structure metrics (SSM) offer an equally general way to measure this gap and to explain the improvements observed across models. As shown in Figure 17, representation alignment helps close the gap between the SSM of a pretrained encoder such as DINOv2 and that of the diffusion features —providing a complementary, spatially-grounded account of why alignment helps.

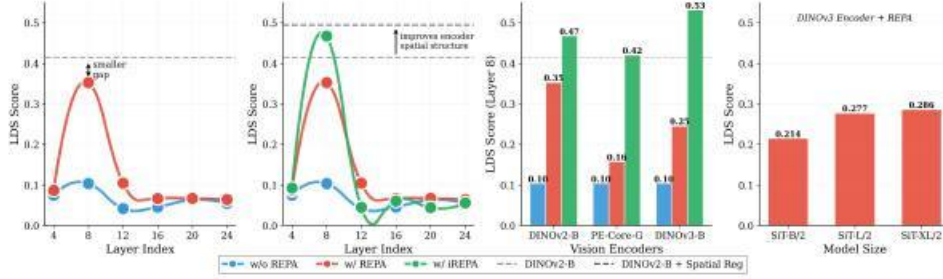


Figure 17: **Representation alignment as bridging the spatial feature gap.** a) Representation alignment can be viewed, more generally, as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder (at the cost of some global information), helping achieve better generation performance. c) iREPA consistently improves performance across a range of vision encoders. d) The spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

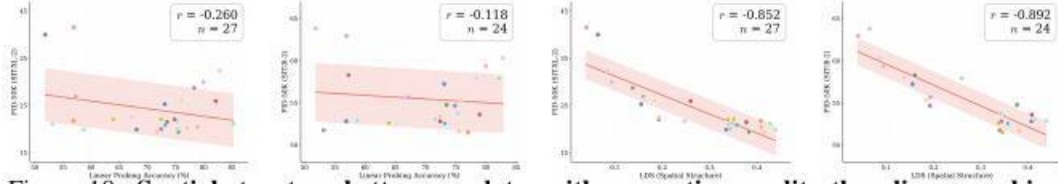


Figure 18: **Spatial structure better correlates with generation quality than linear probing.** Correlation analysis spanning 27 vision encoders and multiple model scales. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) is only weakly correlated. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) is strongly correlated at both scales. See Figure 10 in the Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

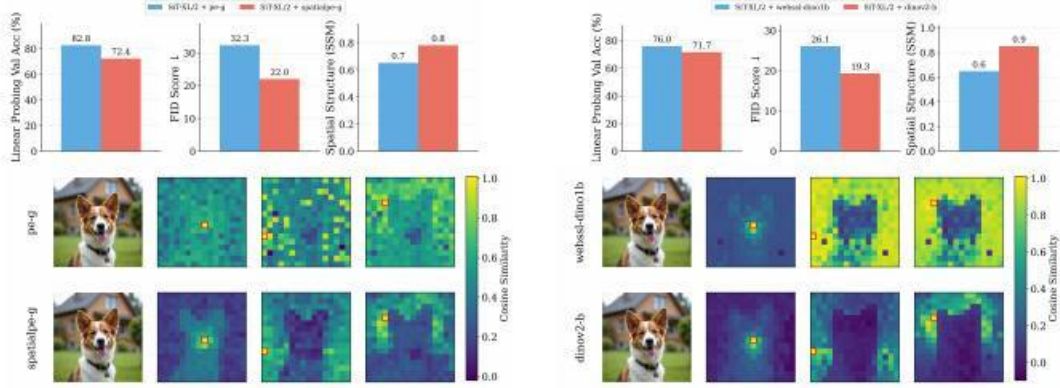


Figure 19: **Motivating examples from recent SSL encoders — spatial structure matters.** **Top row:** Comparisons across encoder families revealing the same inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g attains higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that, across these families, encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

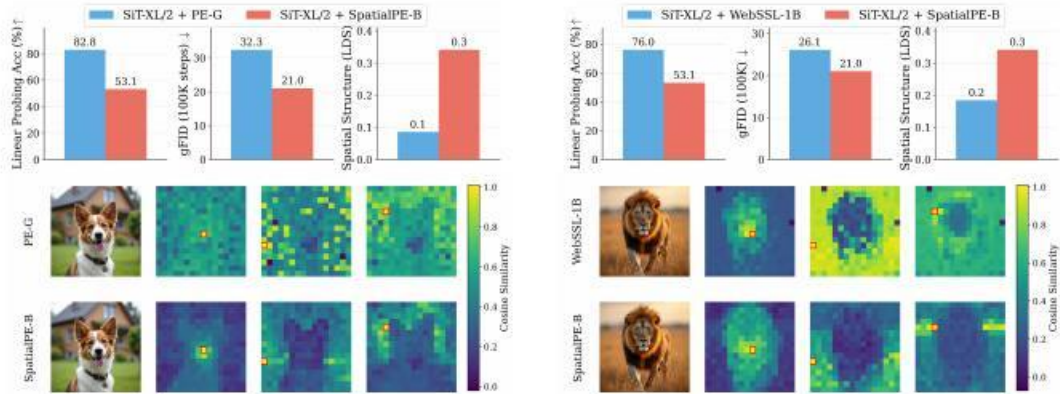


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Comparisons across encoder families, all exhibiting the same inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite a significantly higher validation accuracy (82.8% vs 53.1%), yields worse generation than PE_{Spatial}-B (Bolya et al., 2025). **Right:** Likewise, WebSSL-1B (Fan et al., 2025) has much better global performance (76.0% vs 53.1%) yet worse generation. **Bottom:** Across these families, spatial structure —rather than global performance— provides the better predictor of generation quality. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the impact of spatial normalization on vision encoder representations. Vision encoders commonly carry significant global components or overlays that limit the contrast between spatial tokens; for example, tokens in one semantic region can be highly correlated with tokens in unrelated regions (e.g., background), reducing the spatial distinctiveness of features. Spatial normalization removes this global overlay to enhance contrast between spatial tokens, helping the model better preserve local spatial structure while reducing the dominance of global information that can interfere with generation quality. This behaviour is not encoder-specific but reflects a general tendency of pretrained encoders.

The following figures illustrate this effect across a variety of examples, showing how spatial normalization consistently transforms feature representations to emphasize spatial structure:

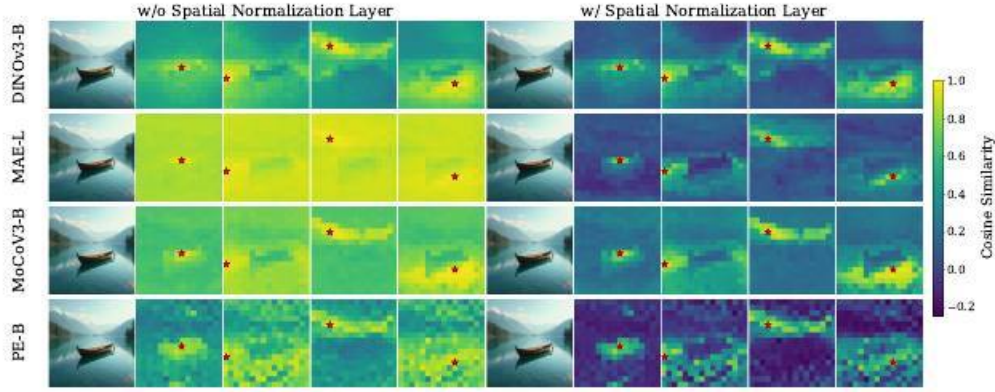


Figure 21: Visualizing the impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while global interference is reduced, yielding more distinct semantic boundaries. This effect is representative of the general behaviour observed across encoders and examples.

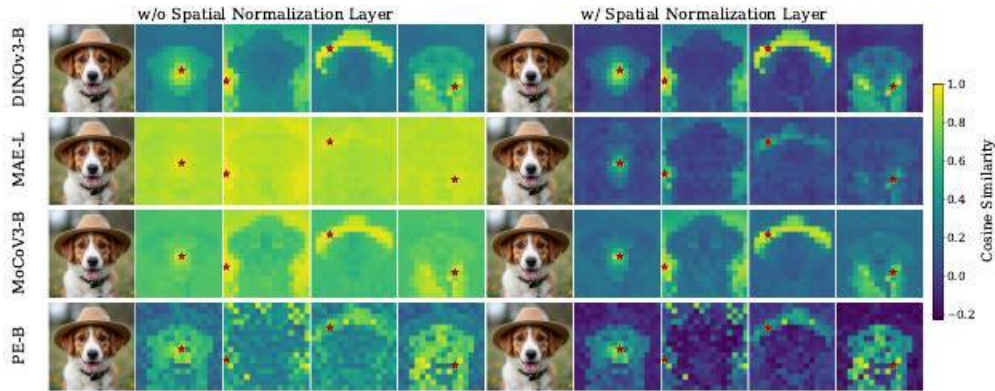


Figure 22: Visualizing the impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while global interference is reduced, yielding more distinct semantic boundaries —as observed consistently across the examples.

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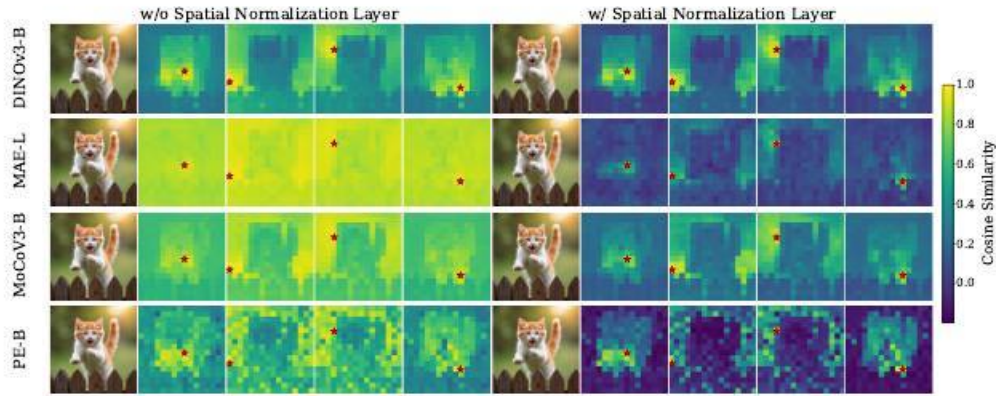


Figure 23: Visualizing the impact of spatial normalization (Example 2). Feature similarity maps illustrate how spatial normalization improves spatial contrast. The original features (left) carry a global overlay that blurs the distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between semantic regions —consistent with the general trend seen across examples.

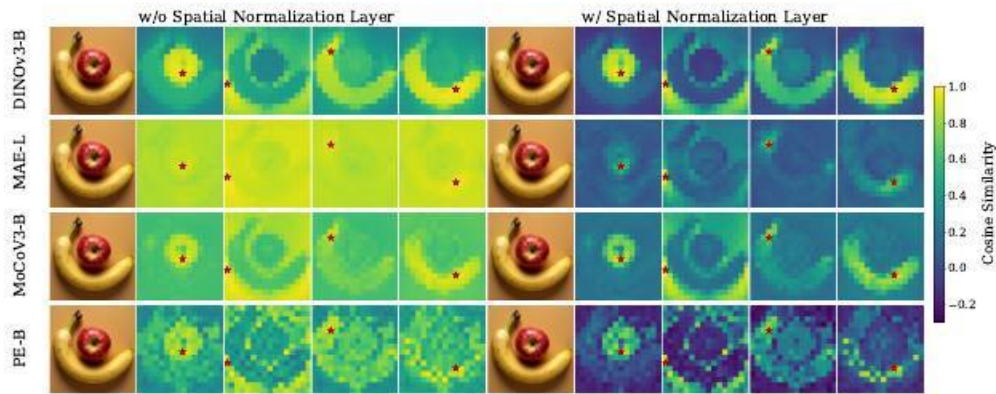


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while global interference is reduced, yielding more distinct semantic boundaries —mirroring the general behaviour seen across the other examples.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. To test whether the proposed spatial improvements extend beyond a single resolution, we conduct further experiments on ImageNet-512 (Deng et al., 2009). We follow the same setup as REPA (Yu et al., 2024) and report results across a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results use SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 shows results with REPA before and after applying the spatial improvements (iREPA). The spatial improvements generalize cleanly to this higher resolution, and the gains remain consistent across the different pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.) —indicating the approach is not tied to the 256×256 regime.

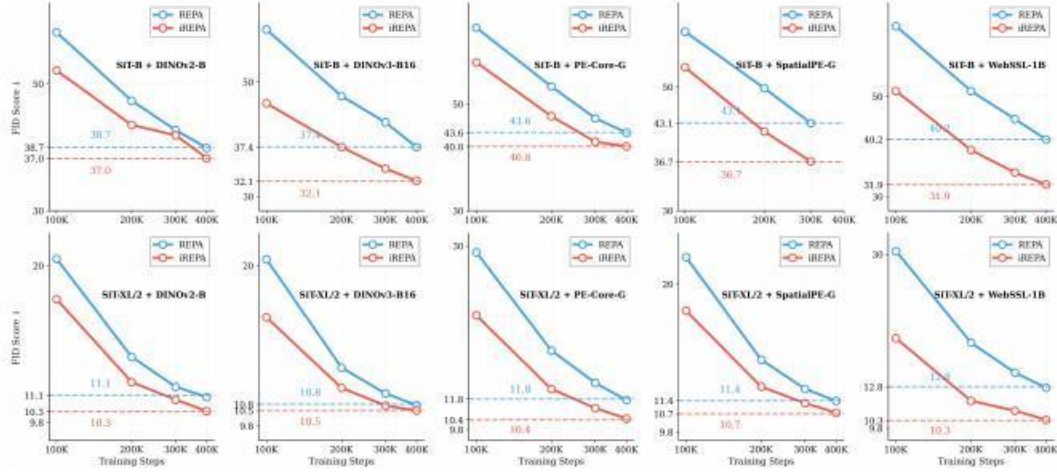


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently improves convergence speed across resolutions, holding for both ImageNet 256 (Figure 7) and ImageNet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To probe the generalizability of the proposed spatial improvements well beyond class-conditional ImageNet generation, we also perform extensive experiments on multimodal text-to-image (T2I) tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a range of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated using an ODE sampler with 50 NFE. Fig. 26 shows results with REPA before and after applying the spatial improvements (iREPA). The spatial improvements carry over to the multimodal T2I setting as well, with consistent gains across the different pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.) —demonstrating that the principle extends across generation tasks and conditioning modalities, not just class-conditional image synthesis.

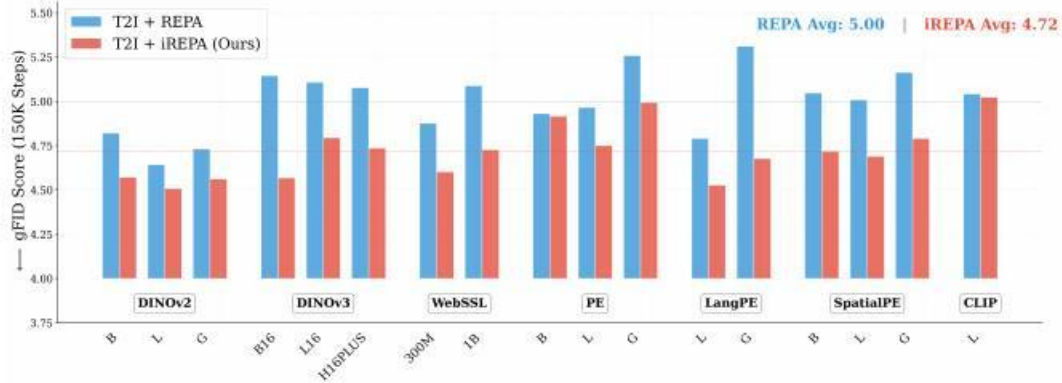


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed in both settings, class-conditional ImageNet (refer §4) and multimodal T2I tasks (above). The gains remain consistent across the different pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.), underscoring the breadth of the effect.

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

To confirm that our conclusions are not driven by a few atypical points, we repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 shows the updated correlations across model sizes. Spatial structure continues to correlate much more strongly with generation performance than linear probing accuracy across all scales. Interestingly, once the outliers are removed, linear probing even exhibits a small *positive* correlation with gFID (i.e., higher linear probing performance is associated with worse generation). This is fully consistent with the general trend discussed in Sec. 2, where target representations with higher global semantic performance (linear probing accuracy) frequently show similar or worse generation performance with representation alignment (REPA).

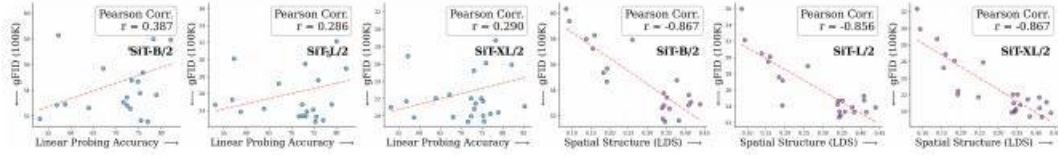


Figure 27: Correlation analysis without outliers. Across all model sizes (B, L, XL), spatial structure continues to correlate much more strongly (Pearson $|r| > 0.85$) with generation performance than linear probing accuracy (Pearson $|r| < 0.38$) even after removing the outliers (MoCOv3-L and MAE-L). Interestingly, once the outliers are removed, linear probing even shows a small positive correlation with gFID (i.e., higher linear probing performance is associated with worse generation). This is consistent with the general trend discussed in Sec. 2, where target representations with higher global semantic performance (linear probing accuracy) frequently show similar or worse generation performance with representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after applying the spatial improvements with REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and a classifier-free guidance scale of 4.0. Mirroring the quantitative gains (§4), the spatial improvements (iREPA) also enhance the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we verify that the proposed findings are robust to the choice of evaluation metric by adopting an alternative measure of generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we use the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. We then re-examine the key findings from both §3 and §4 under this different metric to confirm they are not an artifact of FID.

Spatial structure correlates much higher with generation performance than linear probing. To test the robustness of the analysis from §3, we repeat the large-scale correlation analysis across the same broad set of vision encoders using the CMMD metric (Jayasumana et al., 2024). Results are shown in Figure 29, all reported using SiT-B/2 (100K steps) and REPA. All spatial metrics again show much stronger correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$), confirming that the key empirical findings of §3 generalize across the choice of evaluation metric (FID or CMMD).

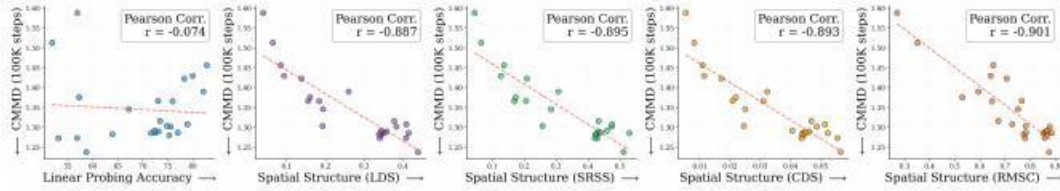


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 using the CMMD metric (Jayasumana et al., 2024) in place of gFID. All results use SiT-B/2 (100K steps) and REPA. All spatial metrics again correlate much more strongly (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$), confirming that the key empirical findings of §3 hold regardless of the evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next assess the generalizability of the analysis from §4 using the CMMD (Jayasumana et al., 2024) metric alongside the traditional evaluation metrics (gFID, sFID, IS, etc.). Results are shown in Table 8. Across the range of vision encoders considered, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, sFID, Prec., Recall). The improvements are consistent both with and without classifier-free guidance (CFG), reinforcing that the benefit is metric- and guidance-agnostic.

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We report additional results using the CMMD metric (Jayasumana et al., 2024). All results use SiT-XL/2, a batch size of 256, and traditional REPA as the baseline. We adopt the PyTorch implementation from (Paul, 2024) for computing CMMD, and use the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. Across the range of vision encoders considered, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, etc.). This confirms that the key empirical findings of §4 hold regardless of the evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features². As shown in Table 9, spatial normalization still helps for SAM2, though the improvement is comparatively marginal. This is expected under our general account: spatial normalization works by removing the global component (mean of patch tokens) to enhance spatial contrast, and since SAM2 already carries little to no global information (validation accuracy < 24%), there is little global component left to remove, so the gains from normalization are small (see Figure 30). This limiting case is thus fully consistent with —rather than an exception to— the broader principle.

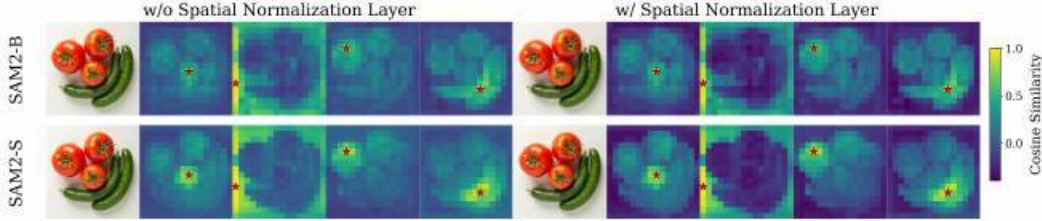


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial normalization layer still helps enhance spatial contrast, though the improvement is comparatively marginal. This behaviour is exactly what the general account predicts: spatial normalization (§4) works by removing the global component (mean of patch tokens) to enhance spatial contrast, and since SAM2 already carries little to no global information (validation accuracy < 24%), there is little to remove, so the improvement in spatial contrast is small.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** Even for this low-global-information encoder, the spatial improvements still help, consistent with the general account. All results use SiT-XL/2, no classifier-free guidance, SAM2-S (46M) as the vision encoder, and traditional REPA as the baseline.

²Note that, as in (Bolya et al., 2025), we use the intermediate output of the vision encoder for SAM2 features rather than the mask logits. As shown in (Bolya et al., 2025), while mask logits yield sharper spatial maps, they are themselves not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. To check that our conclusions do not depend on how global semantic performance is measured, we repeat the correlation analysis from §3 using validation accuracy after full-finetuning³ in place of linear probing. Results are shown in Fig. 31. All results use SiT-XL/2 (100K steps) and REPA. For the full-finetuning validation accuracy, we finetune the entire vision encoder with a learning rate of $5e-5$, a warmup ratio of 0.1, and a total of 30 epochs. All spatial metrics again correlate much more strongly with generation performance than full-finetuning accuracy does. Interestingly, gFID even shows a weak positive correlation with post-finetuning validation accuracy (Pearson $r = 0.317$): as validation accuracy rises, gFID rises and generation quality worsens. This mirrors the general trend from §2, where target representations with higher global semantic performance frequently show similar or worse generation performance with representation alignment (REPA)—confirming the finding holds whether global information is measured by linear probing or full finetuning.

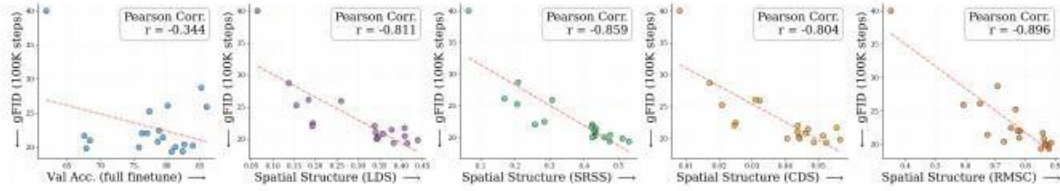


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full-finetuning in place of linear probing. All results use SiT-XL/2 (100K steps) and REPA. All spatial metrics again correlate much more strongly with generation performance than full-finetuning accuracy. Interestingly, gFID even shows a weak positive correlation with post-finetuning validation accuracy (Pearson $r = 0.317$): as validation accuracy rises, gFID rises and generation quality worsens. This mirrors the general trend from §2, where target representations with higher global semantic performance frequently show similar or worse generation performance with representation alignment (REPA).

³Note that while the overall findings are unchanged, in the context of REPA linear probing may be the more faithful estimate of global information in “pretrained” vision encoders, because REPA regularizes with the “pretrained” encoder features themselves and finetuning the encoder can alter the amount of global information. As in REPA (Yu et al., 2024), we therefore adopt linear probing accuracy as our default measure of global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recently proposed state-of-the-art pixel-space diffusion methods, e.g., DeCo (Ma et al., 2025b) —showing the approach remains competitive even outside latent-space diffusion. All results use 200K training iterations with a batch size of 256, evaluated with 50-step Euler sampling and without classifier-free guidance. Results for DeCo are taken directly from (Ma et al., 2025b).